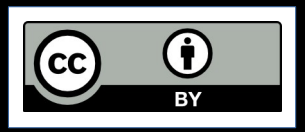
Openstreetmap
(Openwhatevermap?)
Lights and shadows

A well-known crowd-sourcing project in the geospatial world.

Francesco P. Lovergine
CNR IREA
<frankie@linux.com>

...and proud OSMapper since 2008



Immagini estratte da OSM e Wikipedia.
Si vedano i rispettivi copyright.

The big plan

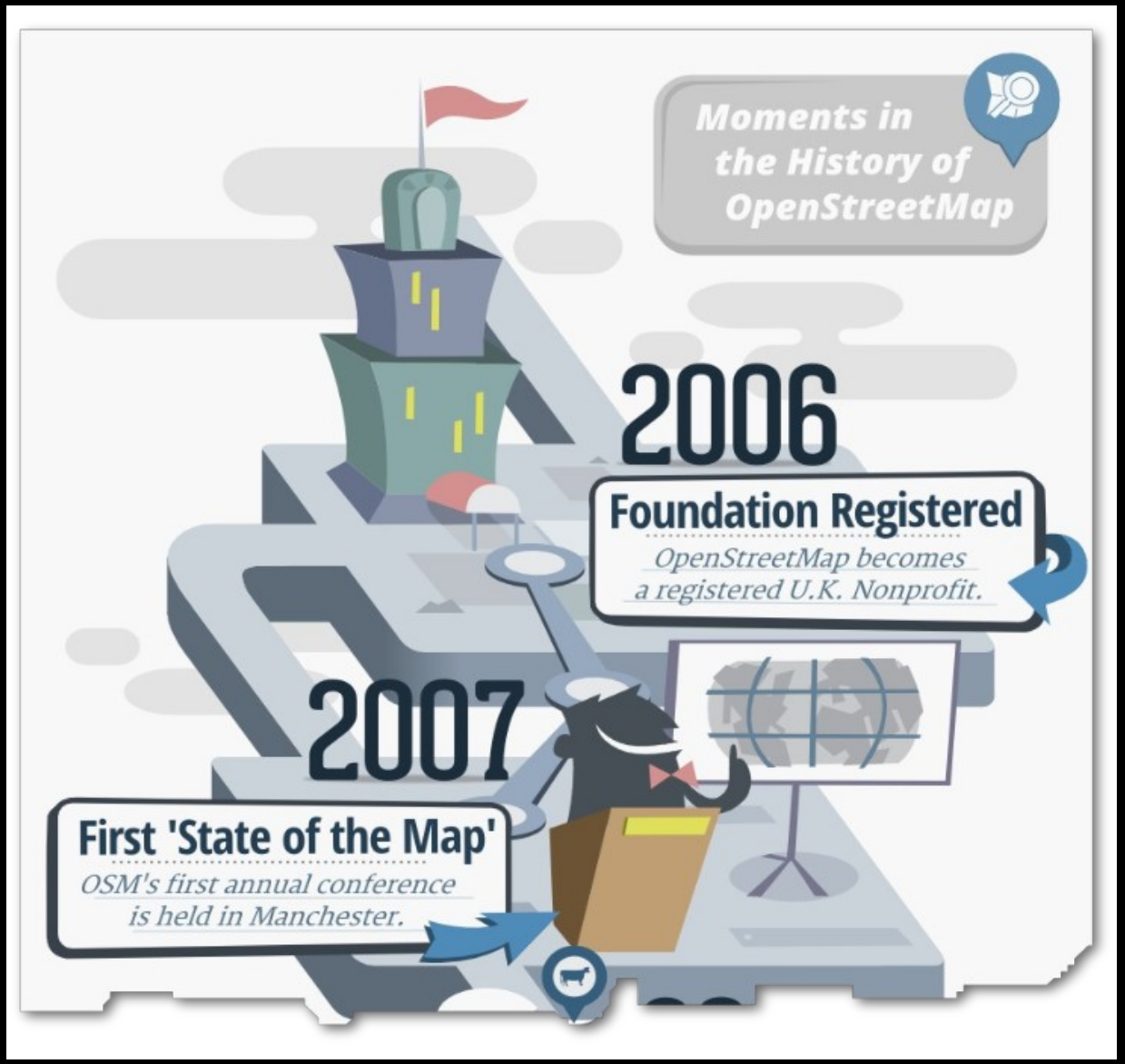
OSM is “The project that creates and distributes *free* geographic data for the world. We started it because most maps you think of as *free* actually have legal or technical restrictions on their use, holding back people from using them in creative, productive, or *unexpected* ways.”

(cit. OSM Wiki)

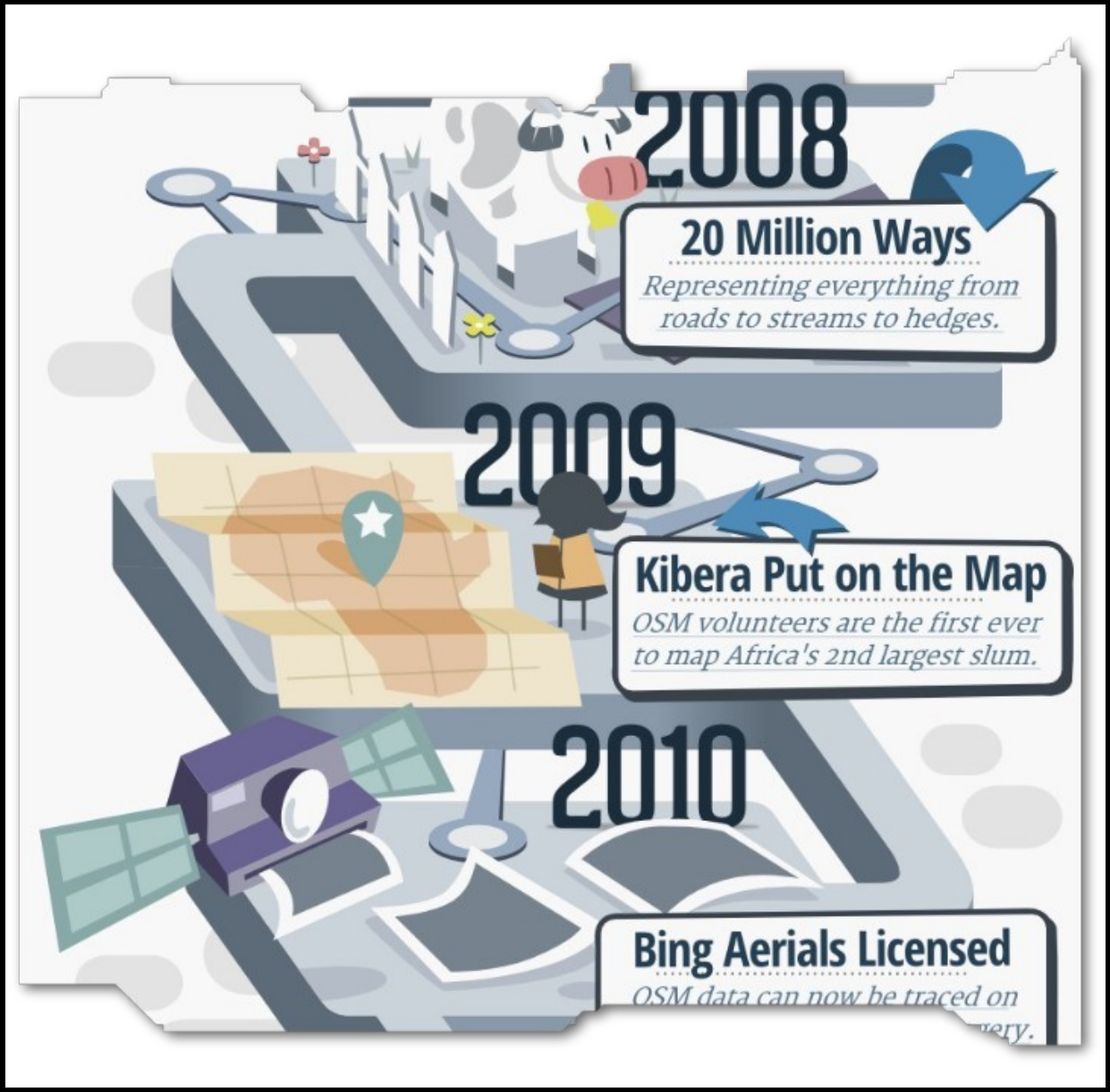
Free as in freedom

…and in the GNU GPL sense

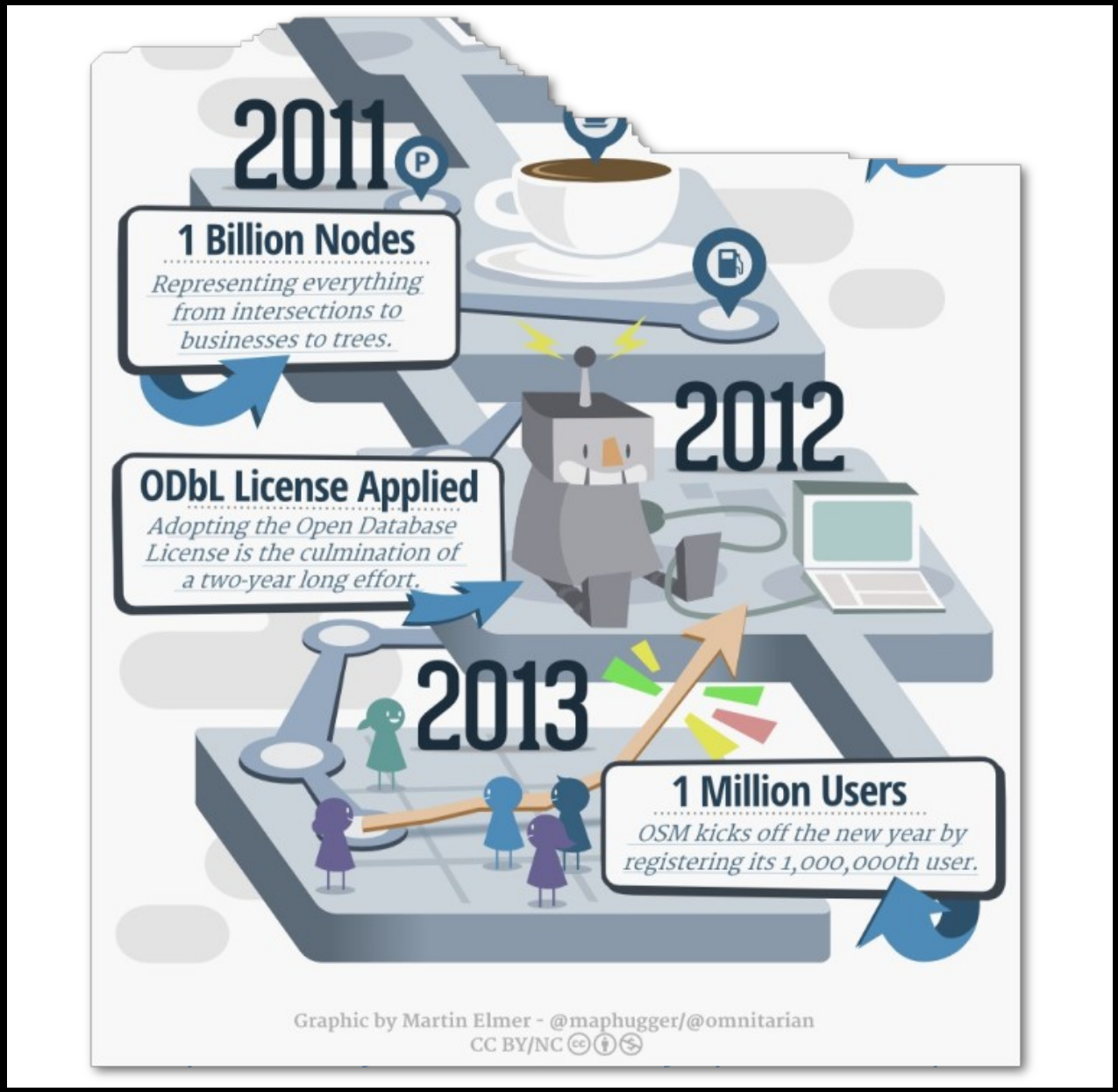
History: birth...



History: ...growth and media blessing...



History: ...a new license and beyond



Today

Copyright © 2011-2015 altogetherlost.com & Pascal Neis (neis-one.org) - Oh U Want More? See ResultMaps-Overview

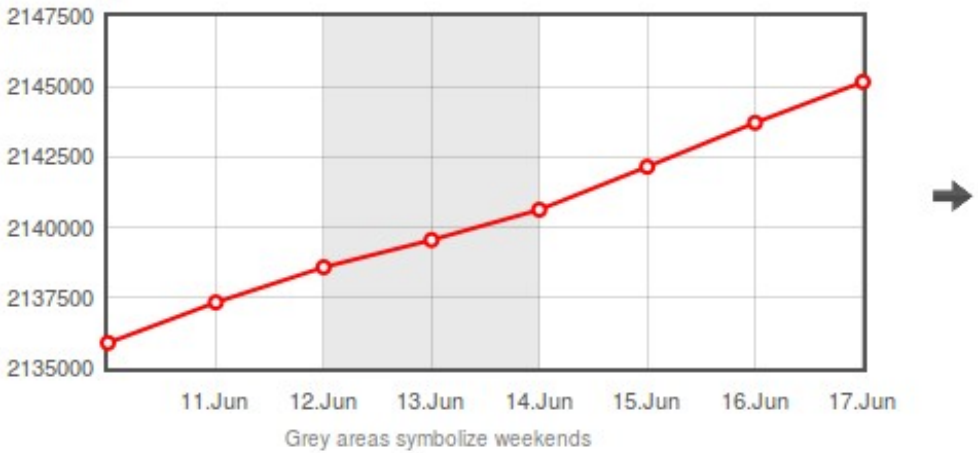
OSMstats **Information about:** Members | Nodes | Ways | Relations | Countries | Changesets **Date:** 1 6 201

Report run at Jun, 18th 2015 1:03 am CET

Statistics of the free wiki world map (OpenStreetMap.org) created in Central European Time (CET)

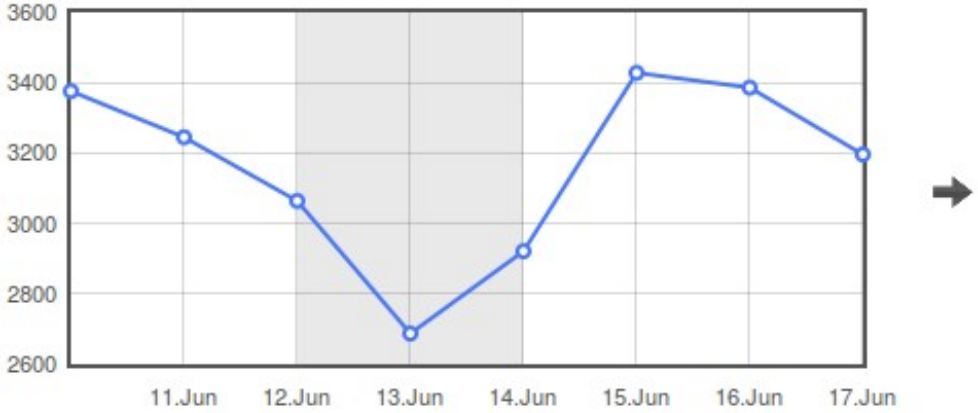
Number of registered OSM members*	2145169
Max. OSM user ID of a member with at least one contribution	2995574
Yesterday's number of active members	3196
Number of active members last 7 days	9785
Number of active members last 30 days	25438
Yesterday's members with >= 15 Node edits	2080
Yesterday's members with >= 5 Way edits	2147
Yesterday's members with >1 Changesets	2096
Number of members who are the last modifier of at least one OSM object (Node, Way or Relation)	396465

No. of registered OSM members



Grey areas symbolize weekends

No. of daily active members overall



Grey areas symbolize weekends

Today

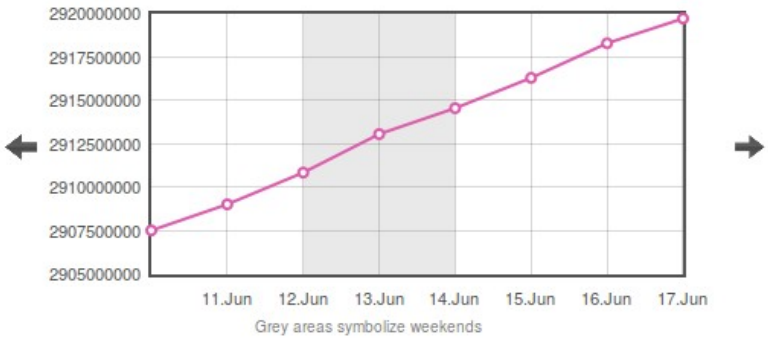
OSMstats **Information about:** Members | Nodes | Ways | Relations | Countries | Changesets **Date:** 1 6 201

Report run at Jun, 18th 2015 1:03 am CET

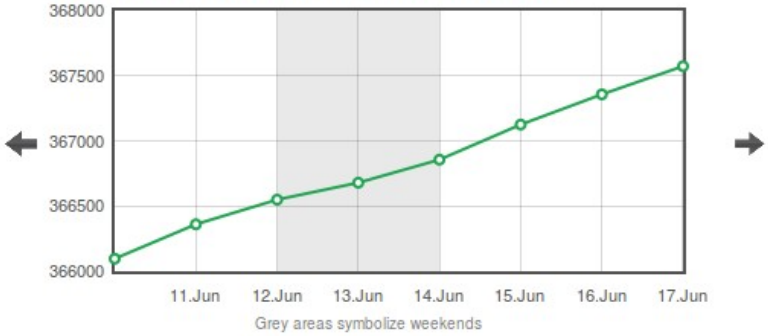
Statistics of the free wiki world map (OpenStreetMap.org) created in Central European Time (CET)

Number of OSM Nodes in the database	2919723075
Number of members who are the last modifier of at least one node	367573
Max. OSM Node ID	3601338433
Yesterday's number of created nodes	1647155
Yesterday's number of modified nodes	282179
Yesterday's number of deleted nodes	209419

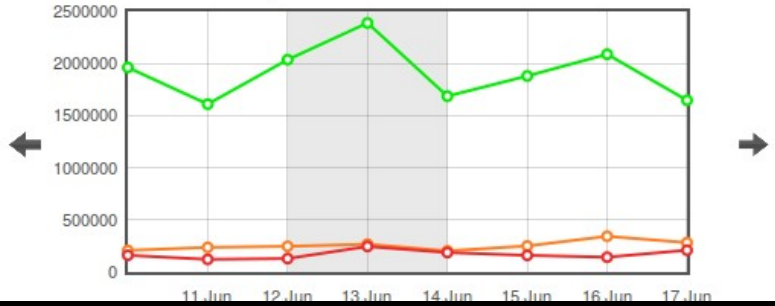
No. of OSM Nodes in the database



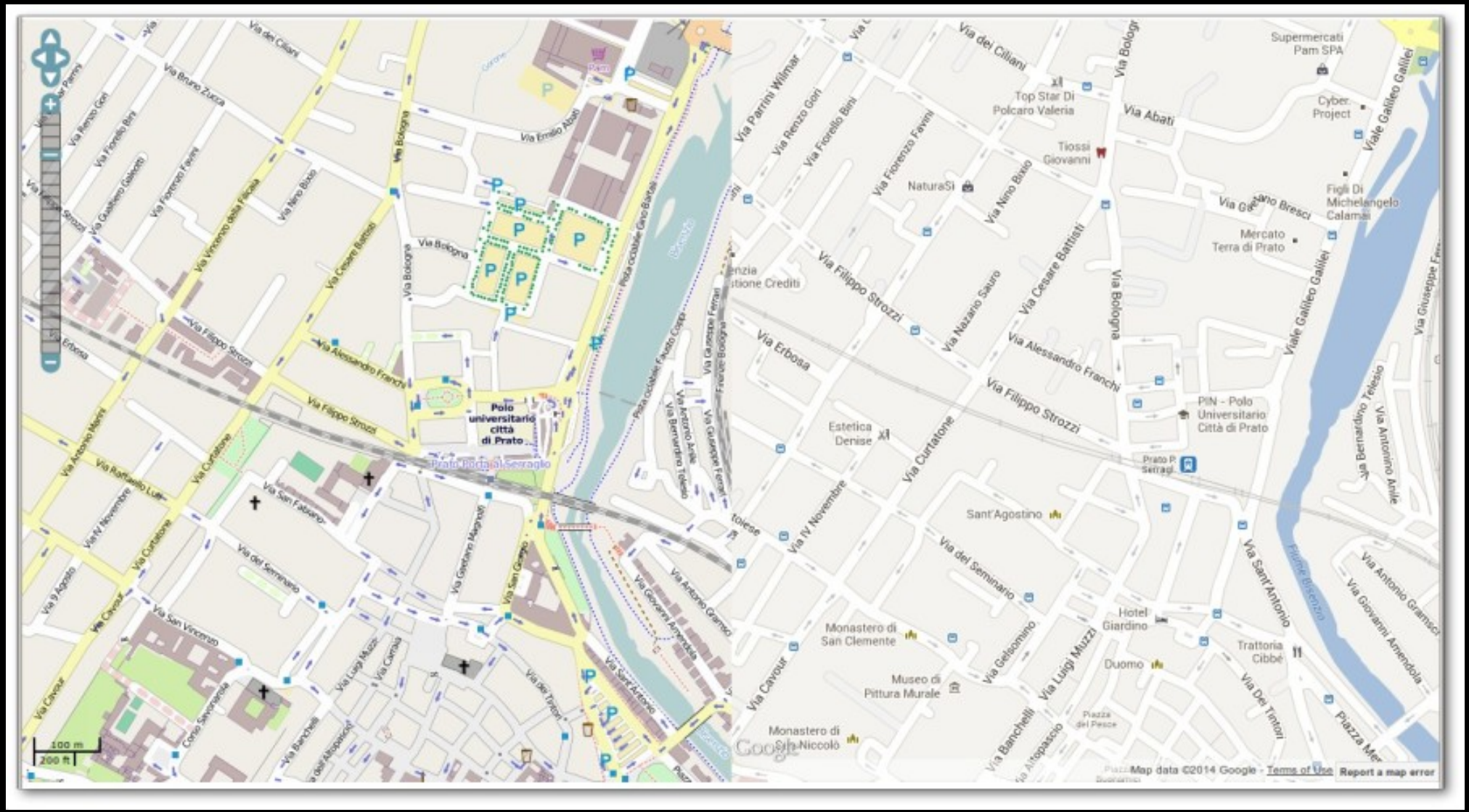
No. of members who are the last modifier of at least one node



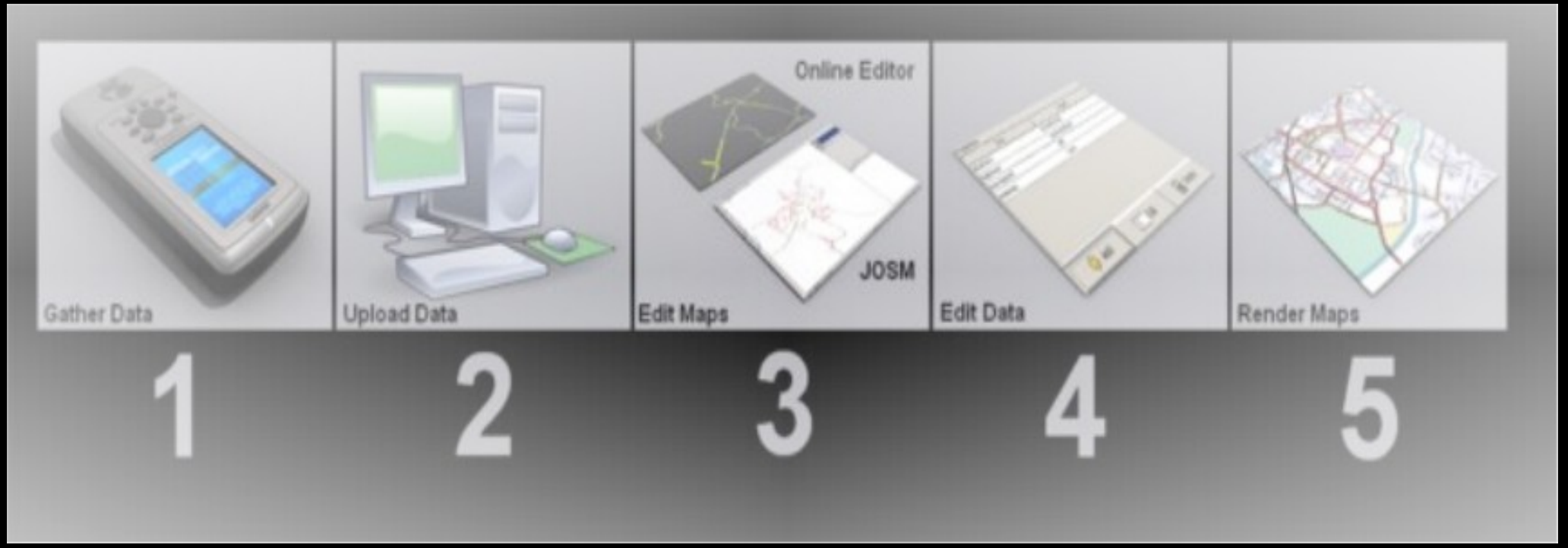
No. of nodes that are daily created/modified/deleted



Unfair comparison: OSM vs GMaps



How to contribute



http://wiki.openstreetmap.org/wiki/Beginners'_guide

OSM Data Model: elements

Data are a combination of:



Node



Way



Relation

OSM Data Model: elements

Ways have multiple meanings:

 **Closed way**

 **Area**

OSM Data Model: tags

Each object has one or more tags which describe its proprierties by means of a KV attribution.

E.g. an area means a parking area if it has a tag

amenity=parking

Where are tags described?

http://wiki.openstreetmap.org/
http://taginfo.openstreetmap.org/

Missing? Just ask or invent!

Features (tags)

Key	Value	Element	Comment	Rendering	Photo
public_transport	platform		The place where passengers are waiting for the transportation vehicle.		
public_transport	station		A station where several buses or trains end their service. Used in combination with `building=yes` or `area=yes`.		
public_transport	stop_position		The position on the street or rails where a bus, tram or train stops.		
public transport	user defined		*All commonly used values according to Taginfo*		

source: http://wiki.openstreetmap.org/wiki/Key:public_transport



node



way



relation

More info:
http://wiki.openstreetmap.org/wiki/Map_Features

http://wiki.openstreetmap.org/wiki/Proposed_features

http://wiki.openstreetmap.org/wiki/Deprecated_features

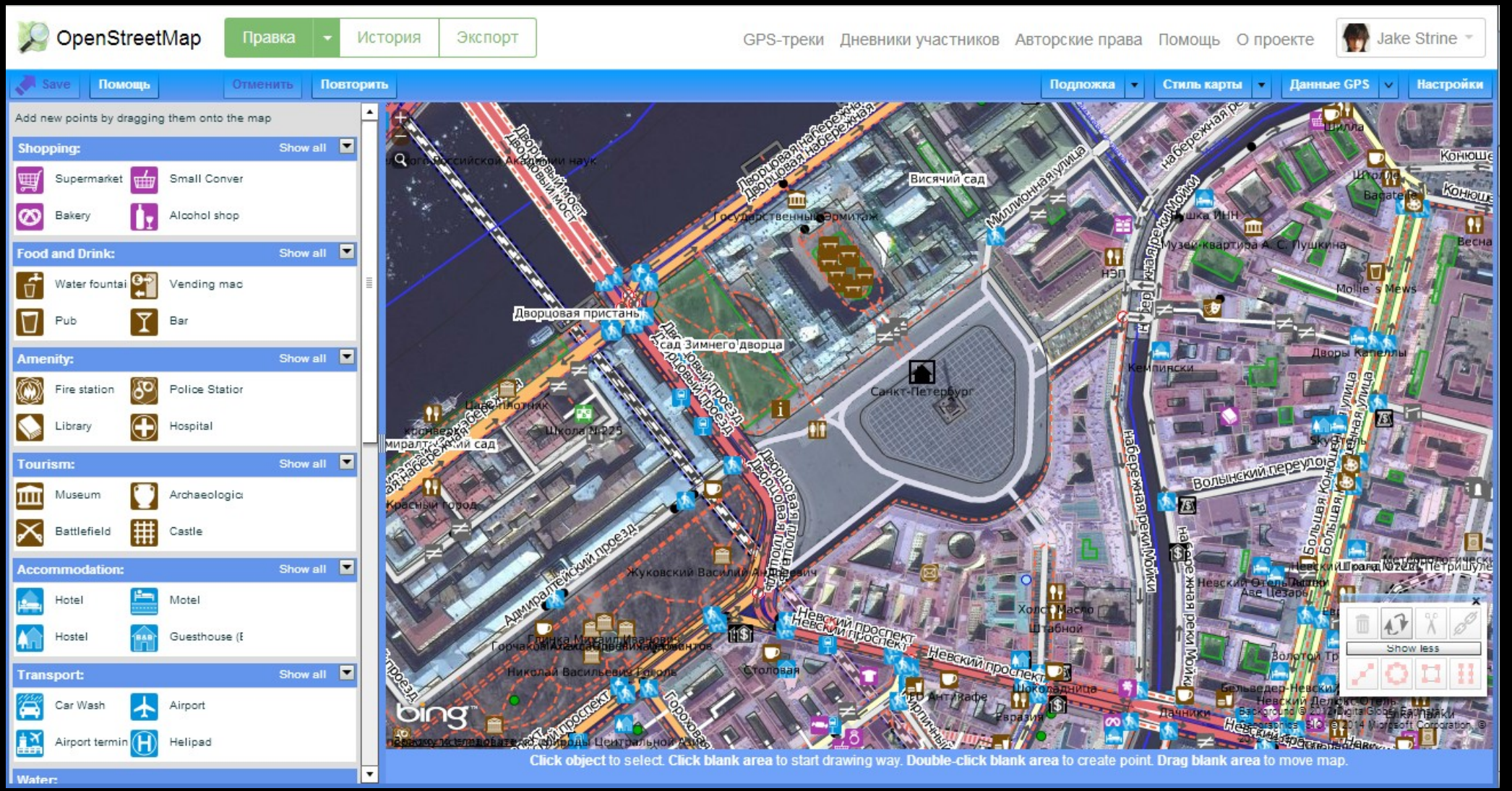
Editing via web clients

The iD editor in-browser editor



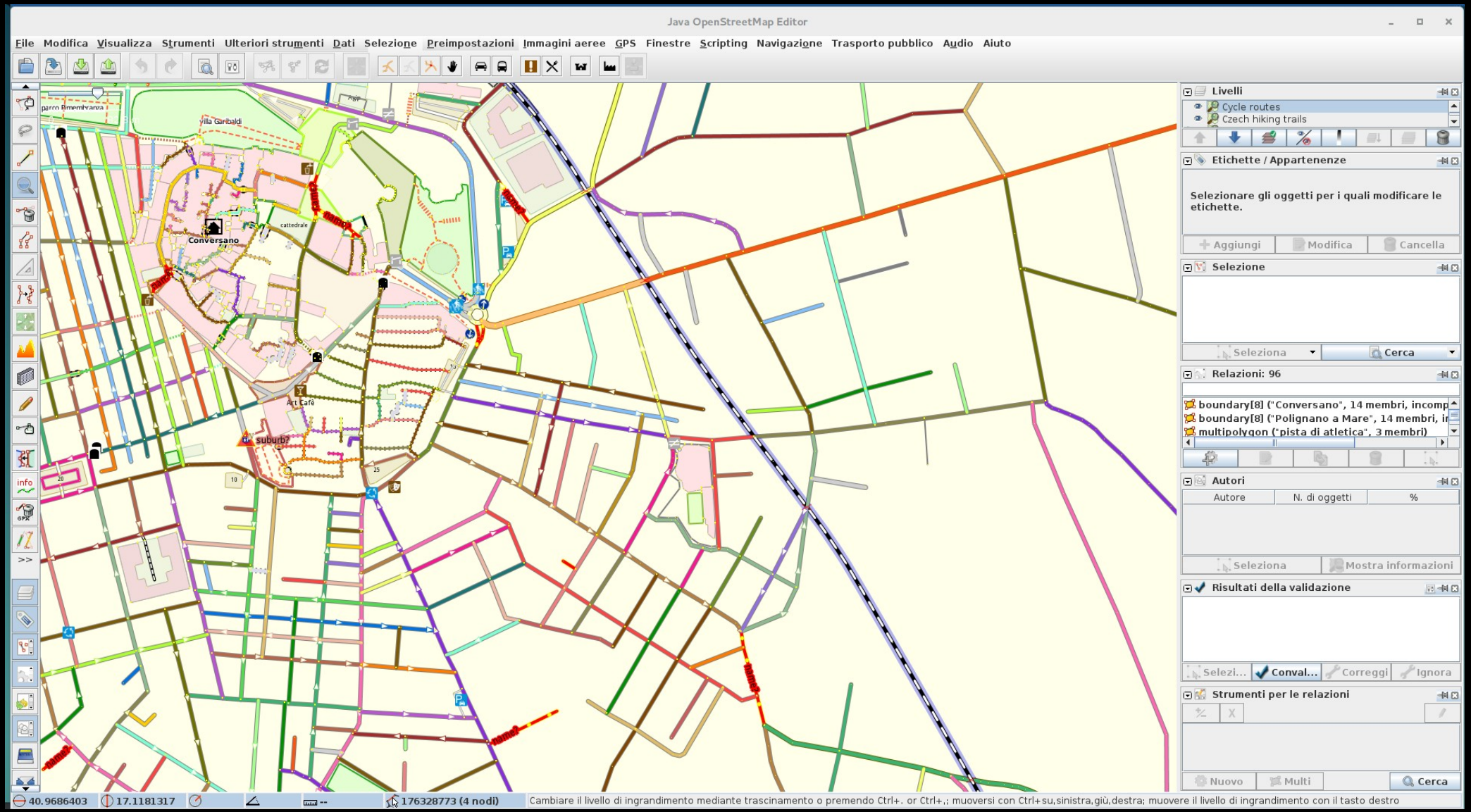
Editing via web clients

The Potlatch web editor



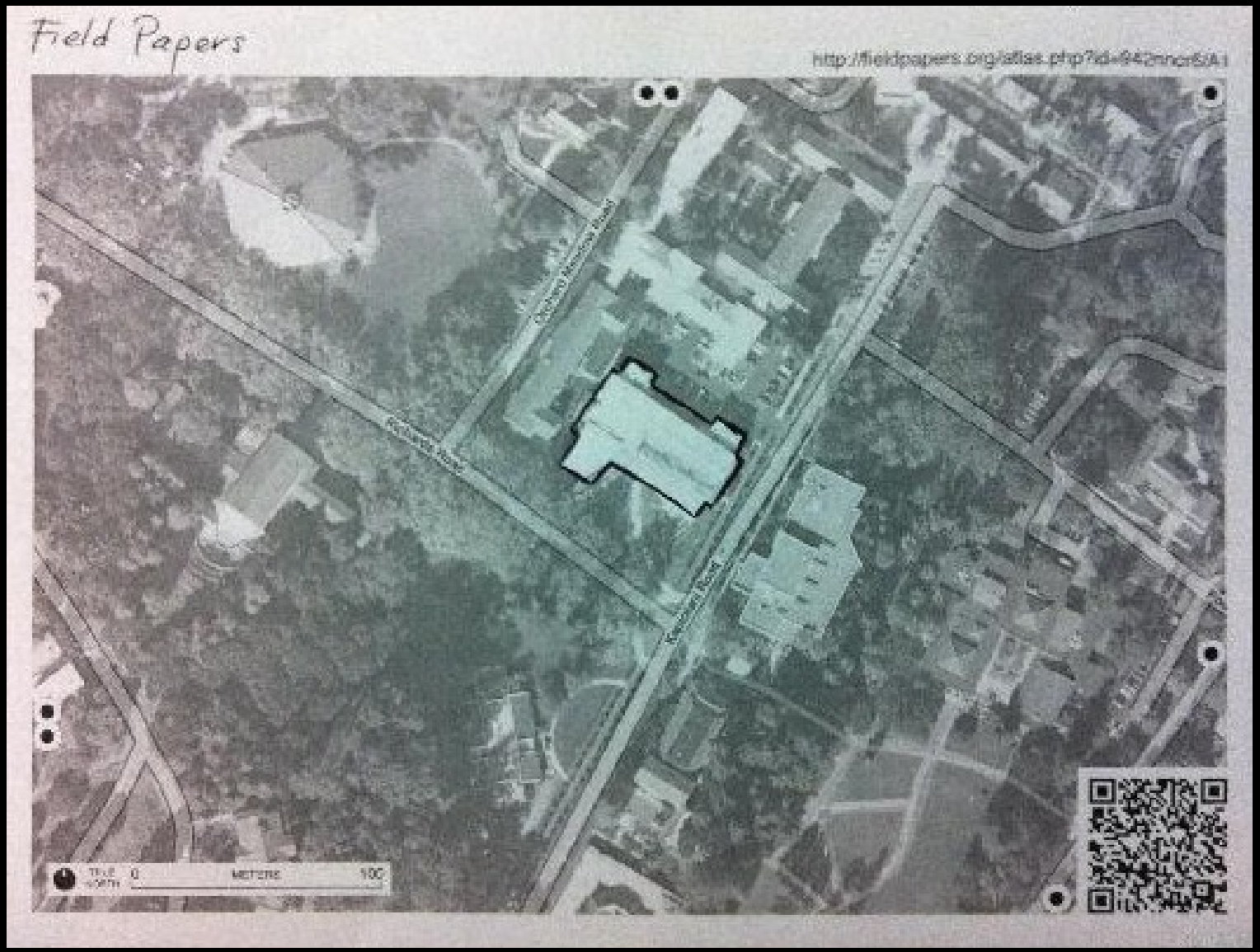
Editing via desktop clients

JOSM

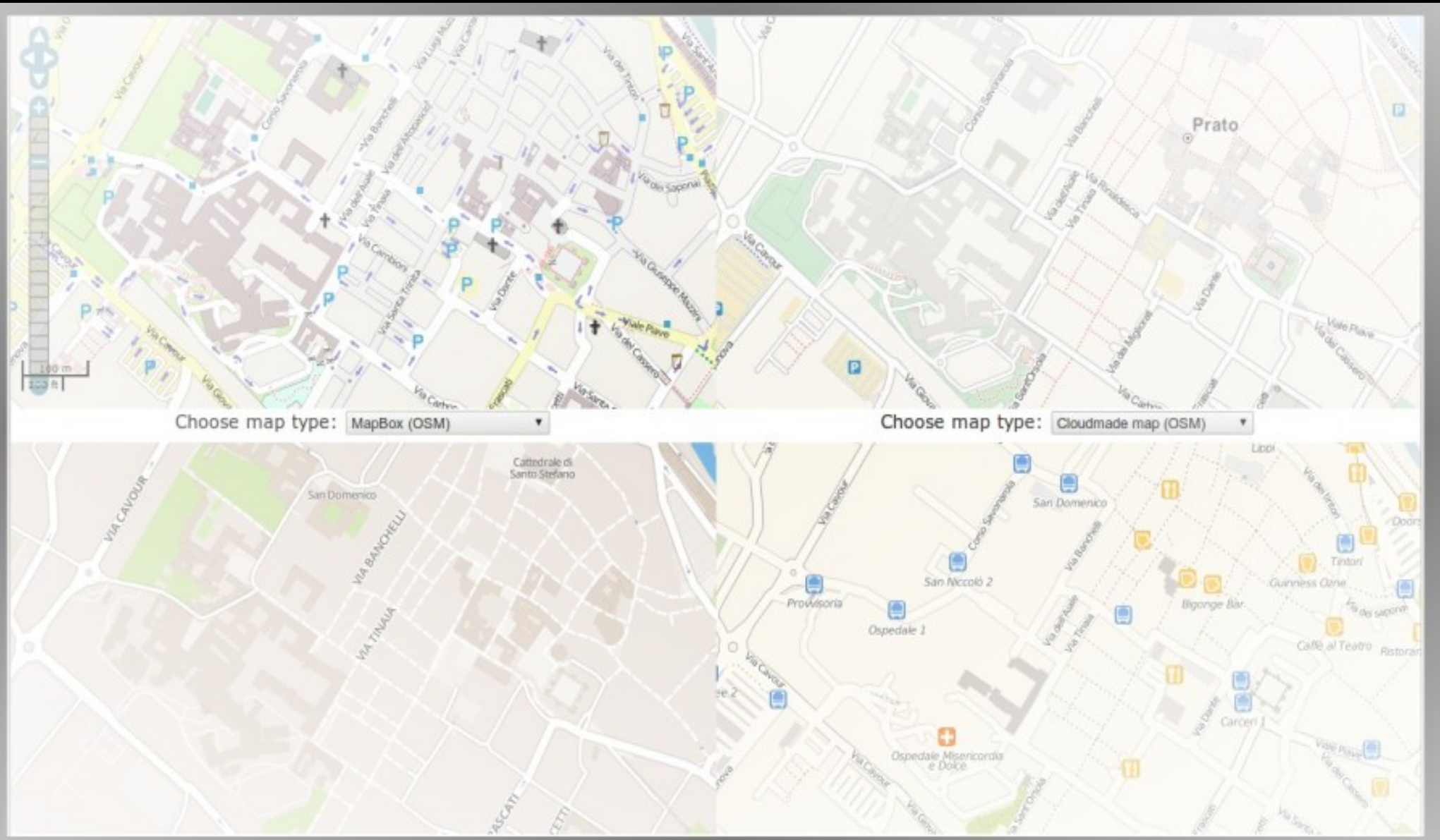


Field job

Field papers or Walking papers



Same data, many renderings



...even unconventional ones!



http://goo.gl/4hj3H5

...and beyond!



A 3D WebGL rendering

Many applications



navfree



orienteering

http://oobrien.com/oom/



osmand



mapOSMatic



4umaps.eu

How to access data

OSM “database” uses XML (plain bz2)/PBF (high-compression) formats
to be used with osmosis or osm2pgsql

~~**43GB/28GB (2015-06-17)**~~
109GB/60GB yesterday (2021-10-22)

http://planet.openstreetmap.org/
(and other mirrors)

Overpass Turbo
National extracts
Local extracts
QGIS plugins

Extracts

- download.geofabrik.de (daily, world)
- geodati.fmach.it (daily, italy)
- WMF OSM toolserver (daily, italy)

Complexity & Flexibility

A great freedom in tags use for OSM
is at the same time the major strength and the major limit of OSM as a geospatial knowledge source

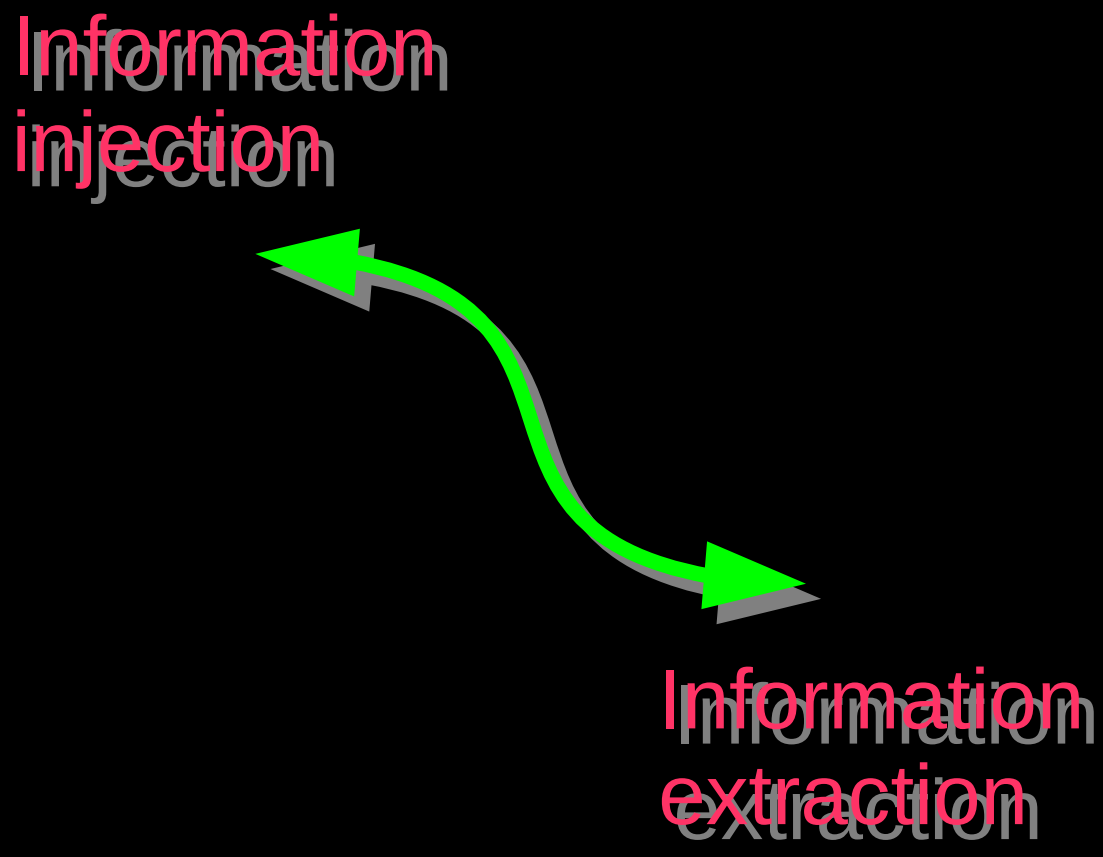
License is currently OdbL+CC-By-SA
http://www.openstreetmap.org/copyright/it
which is a pledge for contributors and anyone seriously interested in OSM use for any purpose.

Geoinformation Ontologies

“The term ontology has its origin in philosophy and has been applied in many different ways. The word element onto- comes from the Greek ὤν, ὄντος, ("being", "that which is"), present participle of the verb εἰμί ("be"). The core meaning within computer science is a model for describing the world that consists of a set of types, properties, and relationship types. There is also generally an expectation that the features of the model in an ontology should closely resemble the real world (related to the object).”

(cit. Wikipedia)

A two ways approach to OSM



Example: land cover and land use

Heterogeneous and often specialistic attributes for local or global areas and their human use

Tags of interest:
natural, landuse, landcover, surface, amenity, place, taxon

Taxonomies:
FAO LCCS, EU Corine, NLCD92

Environmental features:
Power, barrier, cover, treated, …

Mapillary images?

Technical parameters of Corine Land Cover

Technical parameters of Corine Land Cover:[2]

https://wiki.openstreetmap.org/wiki/Corine_Land_Cover

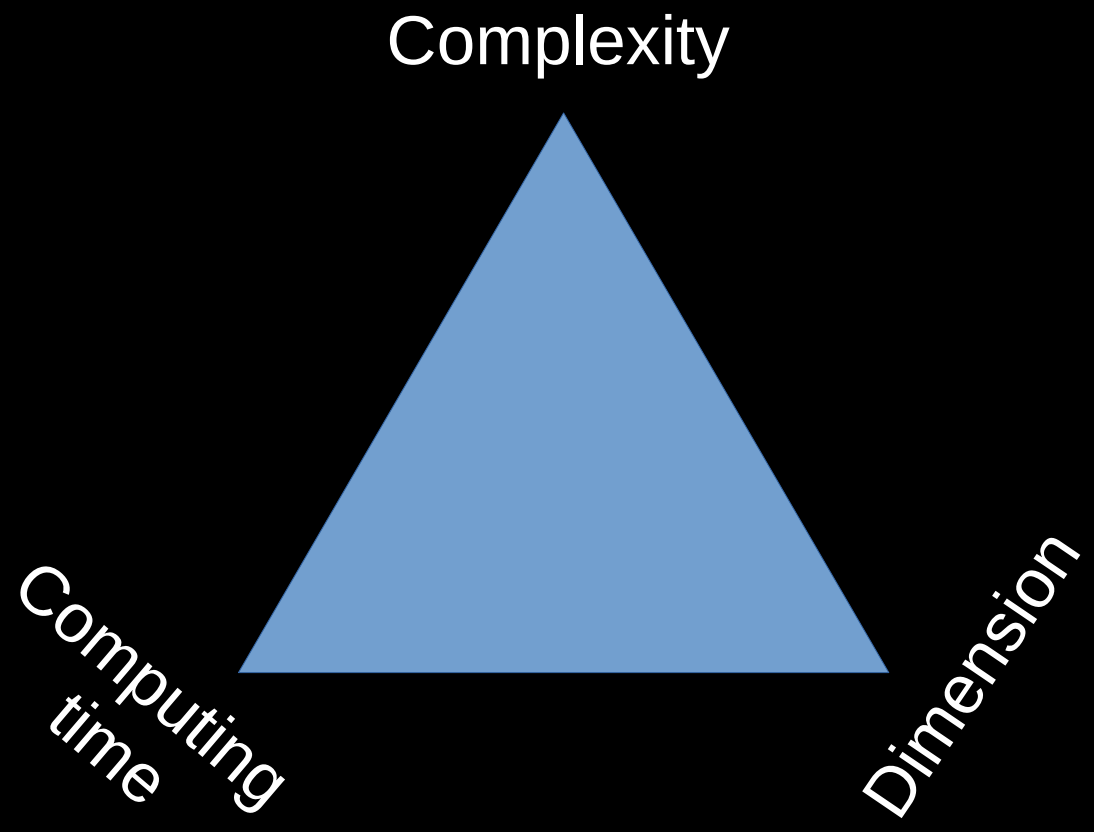
- CLC mapping scale is 1:100.00
- minimum mapping unit (MMU)
- minimum width of linear elements is 100 metres
- MMUS for Land Cover Changes (LCC) since 2000 is 5 hectares.
- 38 countries with a total area of 5.8 Mkm2 participate
- the standard CLC nomenclature includes 44 land cover classes, grouped in a three-level hierarchy. Five main categories are "artificial surfaces", "agricultural areas", "forest and semi-natural areas", "wetlands", "water bodies"

Tagging

The following table details the suggested tag conversion during import. The WikiProject Corine Land Cover/Corine Data Import page gives details of suitable code to do so.

Code	Description	Comment	
1	**Artificial surfaces**		
1.1	**Urban fabric**		
1.1.1	Continuous urban fabric	`landuse=residential`	
1.1.2	Discontinuous urban fabric	`landuse=residential`	
1.2	**Industrial, commercial and transport units**		
1.2.1	Industrial or commercial units	`landuse=industrial;retail` + `note=CLC import: update the tag landuse with either industrial or retail after survey`	
1.2.2	Road and rail networks and associated land	No compatible tag found (mix road and rail landuse; landuse=highway doesn't exist)	
1.2.3	Port areas	`landuse=harbour` Can be replaced by `leisure=marina` after survey	
1.2.4	Airports	`aeroway=aerodrome`	
1.3	**Mine, dump and construction sites**		
1.3.1	Mineral extraction sites	`landuse=quarry`	
1.3.2	Dump sites	`landuse=landfill`	
1.3.3	Construction sites	`landuse=construction`	
1.4	**Artificial, non-agricultural vegetated areas**		

Dive in the Big Data paradigm through OSM!



What OSM is today

Most applications are dedicated to:

- Roping
- Rendering
- Routing

True potential of OSM as as a knowledge base still needs to be unearthed

?

Items for research and deepening

- Rendering of thematic maps
- Massive importing of geospatial data
- Ontologies eand taxonomies (dictionaries) for geology, ecology, environment studies
- Validation and QA
- Machine learning applied to OSM tags for geocoding and knowledge-base extraction or advanced searching
- Big data technologies applied to OSM data (Mapreduce?)